Duong Quoc Hoang

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CAREER OBJECTIVE

Computer Science student at PTIT (GPA: 3.67/4.0, merit scholarship for 3 consecutive semesters) who builds products end to end — Solidity contracts, NestJS services and React frontends. Based in Hanoi (Ha Dong area), available onsite 20+ hours/week. Seeking a Fullstack Software Engineer Intern position in Blockchain and AI products.

PROJECTS

3D GameFi NFT Marketplace & Custom On-chain Indexer **2026**

Solidity, Hardhat, Node.js, Viem, MongoDB, React, Three.js / React Three Fiber, IPFS

- Built a decentralized 3D game-item marketplace in three layers: 2 Solidity contracts (ERC-721 minting and a reentrancy-guarded escrow marketplace), a Node.js indexer, and a Web3 frontend.
- Wrote a custom on-chain event indexer instead of using The Graph: batched block scanning with Viem, persistent sync checkpoints in MongoDB, and idempotent upserts that removed duplicate-key conflicts on replayed events.
- Designed a 3-gateway IPFS fallback chain (Pinata, dweb.link, Cloudflare) so 3D assets still resolve when a primary gateway was unavailable.
- Built the Web3 client in React: wallet connection and typed contract calls through Viem, in-browser .glb rendering with React Three Fiber, and UI flows for minting, listing and buying items.

IELTS Practice Test Platform — Backend (In Progress) **2026**

NestJS, TypeScript, PostgreSQL, Prisma, Redis, Passport-JWT, Jest

- Modeled the domain in Prisma / PostgreSQL: 7 entities spanning classes, enrollment, tests, questions, submissions and attendance, with composite indexes, cascade rules and uniqueness constraints; MCQ, gap-fill and True/False/Not-Given share one JSON answer-key schema.
- Implemented a 5-endpoint auth module with refresh-token rotation: tokens are SHA-256 hashed before storage in Redis and revoked on reuse, with separate access/refresh JWT strategies and bcrypt.
- Built email verification with Redis-backed OTP and Handlebars templates via @nestjs-modules/mailer; validated all request payloads with DTOs and class-validator.
- Unit tested the auth service with Jest, mocking the Prisma, Redis and mail dependencies; next up is the homework module (test, question and submission APIs with auto-grading).

ALGORITHMS & COMPETITIVE PROGRAMMING

- Solved 300+ algorithmic problems in C++ covering graph theory, dynamic programming, greedy, binary search, monotonic stack and meet-in-the-middle.
- Completed the Samsung Applied Algorithm Course (algorithm design under time pressure).

SKILLS & KNOWLEDGE

Languages	TypeScript, JavaScript, C++, Solidity.
Backend	NestJS, Node.js, Prisma, Redis, REST APIs, JWT / Passport, Jest.
Frontend	React, Vite, Three.js / React Three Fiber, Viem (wallet & contract calls).
Database	PostgreSQL, MongoDB.
Blockchain	Solidity, Hardhat, Viem, ERC-721, IPFS, on-chain event indexing.
Tools	Git, Docker, Linux (WSL).

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT) **2024 – 2028**

Bachelor of Computer Science **GPA: 3.67 / 4.0**

Merit-based Academic Scholarship — 3 consecutive semesters

CERTIFICATIONS

Samsung Applied Algorithm Course Certificate August 2025

TOEIC 760 (English) September 2025